

Ratio, Root, and Alternating Series Tests

ANSWER KEY



The Ratio Test

- 1 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2} < 1. \text{ Converges by the Ratio Test.}$$

- 2 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ converges.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)!/3^{n+1}}{n!/3^n} = \lim_{n \rightarrow \infty} \frac{n+1}{3} = \infty > 1. \text{ Diverges by the Ratio Test.}$$

The Root Test

- 3 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n$ converges.

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{n}{2n+1} \right)^n \right|} = \lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2} < 1. \text{ Converges by the Root Test.}$$

Alternating series and the Alternating Series Test

- 4 State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

The AST states $\sum (-1)^{n+1} b_n$ converges if $b_n > 0$, b_n is decreasing, and $\lim_{n \rightarrow \infty} b_n = 0$. Here $b_n = \frac{1}{n}$: positive, decreasing, and $\rightarrow 0$. The series converges (this is the alternating harmonic series).

- 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

$b_n = \frac{n}{n+1} \rightarrow 1 \neq 0$. Since b_n does not approach 0, the Alternating Series Test does not apply, and in fact the series diverges by the n th-term test.

Alternating series error bound

- 6 Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.

$$S_4 = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} \approx 1 - 0.25 + 0.111 - 0.0625 = 0.7986. \text{ Error bound: } |S - S_4| \leq b_5 = \frac{1}{25} = 0.04.$$

Absolute vs. conditional convergence

- 7 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.

$$\sum \left| \frac{(-1)^{n+1}}{n^2} \right| = \sum \frac{1}{n^2}, \text{ a convergent p-series } (p = 2 > 1). \text{ So the series converges ABSOLUTELY.}$$

- 8 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges absolutely, conditionally, or diverges.

$\sum \left| \frac{(-1)^{n+1}}{n} \right| = \sum \frac{1}{n}$, the harmonic series, which diverges. But the alternating series itself converges (shown earlier by the AST). So it converges **CONDITIONALLY**.

More Ratio Test practice

- 9 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{3^{n+1}/(n+1)!}{3^n/n!} = \lim_{n \rightarrow \infty} \frac{3}{n+1} = 0 < 1$. Converges by the Ratio Test.

- 10 Explain why the Ratio Test is **INCONCLUSIVE** for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$. Since the limit equals exactly 1, the Ratio Test gives no conclusion. The p-series test (or Integral Test) should be used instead, correctly showing divergence.

Partial sums of an alternating series (visualizing oscillating convergence)

- 11 Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

$S_1 = 1, S_2 = 1 - \frac{1}{2} = 0.5, S_3 = 0.5 + \frac{1}{3} \approx 0.833, S_4 \approx 0.833 - 0.25 = 0.583, S_5 \approx 0.583 + 0.2 = 0.783$. The partial sums oscillate above and below the true sum $\ln 2 \approx 0.693$, with each successive sum landing closer — a signature behavior of alternating series.

Determining the number of terms needed for a given accuracy

- 12 For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

Need $b_{n+1} = \frac{1}{(n+1)^3} < 0.001 \Rightarrow (n+1)^3 > 1000 \Rightarrow n+1 > 10 \Rightarrow n \geq 10$. So 10 terms suffice.

Mixed practice: classify each series

- 13 Classify each series as convergent (state absolutely or conditionally) or divergent:

a $\sum \frac{(-1)^n}{\sqrt{n}}$ Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \rightarrow 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$)

b $\sum \frac{(-1)^n n^2}{n^3 + 1}$ Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3 + 1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$

c $\sum \frac{(-1)^n}{n!}$ Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\rightarrow 0$)

The Root Test with more complex expressions

- 14 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$ converges.

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2n+1}{3n-1}\right)^n} = \lim_{n \rightarrow \infty} \frac{2n+1}{3n-1} = \frac{2}{3} < 1. \text{ Converges by the Root Test.}$$

- 15 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$ converges.

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{2^n}{n^n}} = \lim_{n \rightarrow \infty} \frac{2}{n} = 0 < 1. \text{ Converges by the Root Test.}$$

Why the Ratio Test fails for p-series

- 16 Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p .

$\lim_{n \rightarrow \infty} \left| \frac{1/(n+1)^p}{1/n^p} \right| = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^p = 1^p = 1$ for ANY value of p . Since the limit always equals exactly 1, the Ratio Test can never determine convergence or divergence for p-series — a different test (the p-series test itself) is required.

Combining multiple tests in one problem

- 17 Determine whether $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$ converges absolutely, conditionally, or diverges.

Test absolute convergence with the Ratio Test on $\left| \frac{(-2)^n}{n^2} \right| = \frac{2^n}{n^2}$: $\lim_{n \rightarrow \infty} \left| \frac{2^{n+1}/(n+1)^2}{2^n/n^2} \right| = \lim_{n \rightarrow \infty} \frac{2n^2}{(n+1)^2} = 2 > 1$. Diverges (even in absolute value) — so the original series DIVERGES (the terms don't even approach 0 since $|a_n| \rightarrow \infty$).

Multiple choice — easy 15

- 18 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{3^n}$.

a. 3

b. $\frac{1}{3}$ ✓

c. $\frac{4}{3}$

d. $\frac{1}{4}$

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/3^{n+1}}{n/3^n} = \frac{n+1}{3n} \rightarrow \frac{1}{3}$. Since $\frac{1}{3} < 1$, the series converges by the Ratio Test.

19 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{8^n}$.

a. 8

b. $\frac{9}{8}$

c. $\frac{1}{8}$ ✓

d. $\frac{1}{9}$

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/8^{n+1}}{n/8^n} = \frac{n+1}{8n} \rightarrow \frac{1}{8}$. Since $\frac{1}{8} < 1$, the series converges by the Ratio Test.

20 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{14^n}$.

a. 14

b. $\frac{1}{14}$ ✓

c. $\frac{1}{15}$

d. $\frac{15}{14}$

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/14^{n+1}}{n/14^n} = \frac{n+1}{14n} \rightarrow \frac{1}{14}$. Since $\frac{1}{14} < 1$, the series converges by the Ratio Test.

21 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{15^n}$.

a. 15

b. $\frac{1}{16}$

c. $\frac{16}{15}$

d. $\frac{1}{15}$ ✓

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/15^{n+1}}{n/15^n} = \frac{n+1}{15n} \rightarrow \frac{1}{15}$. Since $\frac{1}{15} < 1$, the series converges by the Ratio Test.

22 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{10^n}$.

a. 10

b. $\frac{1}{11}$

c. $\frac{1}{10}$ ✓

d. $\frac{11}{10}$

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/10^{n+1}}{n/10^n} = \frac{n+1}{10n} \rightarrow \frac{1}{10}$. Since $\frac{1}{10} < 1$, the series converges by the Ratio Test.

23 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{19^n}$.

a. $\frac{1}{20}$

b. $\frac{20}{19}$

c. $\frac{1}{19}$ ✓

d. 19

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/19^{n+1}}{n/19^n} = \frac{n+1}{19n} \rightarrow \frac{1}{19}$. Since $\frac{1}{19} < 1$, the series converges by the Ratio Test.

24 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{13^n}$.

a. $\frac{1}{14}$

b. 13

c. $\frac{14}{13}$

d. $\frac{1}{13}$ ✓

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/13^{n+1}}{n/13^n} = \frac{n+1}{13n} \rightarrow \frac{1}{13}$. Since $\frac{1}{13} < 1$, the series converges by the Ratio Test.

25 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{5^n}$.

a. $\frac{6}{5}$

b. 5

c. $\frac{1}{6}$

d. $\frac{1}{5}$ ✓

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/5^{n+1}}{n/5^n} = \frac{n+1}{5n} \rightarrow \frac{1}{5}$. Since $\frac{1}{5} < 1$, the series converges by the Ratio Test.

26 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{11^n}$.

a. $\frac{1}{11}$ ✓

b. $\frac{1}{12}$

c. 11

d. $\frac{12}{11}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/11^{n+1}}{n/11^n} = \frac{n+1}{11n} \rightarrow \frac{1}{11}. \text{ Since } \frac{1}{11} < 1, \text{ the series converges by the Ratio Test.}$$

27 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{20^n}$.

a. $\frac{1}{21}$

b. 20

c. $\frac{21}{20}$

d. $\frac{1}{20}$ ✓

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/20^{n+1}}{n/20^n} = \frac{n+1}{20n} \rightarrow \frac{1}{20}. \text{ Since } \frac{1}{20} < 1, \text{ the series converges by the Ratio Test.}$$

28 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{4^n}$.

a. $\frac{5}{4}$

b. 4

c. $\frac{1}{5}$

d. $\frac{1}{4}$ ✓

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/4^{n+1}}{n/4^n} = \frac{n+1}{4n} \rightarrow \frac{1}{4}. \text{ Since } \frac{1}{4} < 1, \text{ the series converges by the Ratio Test.}$$

29 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{12^n}$.

a. $\frac{13}{12}$

b. $\frac{1}{13}$

c. $\frac{1}{12}$ ✓

d. 12

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/12^{n+1}}{n/12^n} = \frac{n+1}{12n} \rightarrow \frac{1}{12}. \text{ Since } \frac{1}{12} < 1, \text{ the series converges by the Ratio Test.}$$

30 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{17^n}$.

a. $\frac{18}{17}$

b. $\frac{1}{18}$

c. $\frac{1}{17}$ ✓

d. 17

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/17^{n+1}}{n/17^n} = \frac{n+1}{17n} \rightarrow \frac{1}{17}$. Since $\frac{1}{17} < 1$, the series converges by the Ratio Test.

31 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{16^n}$.

a. $\frac{1}{16}$ ✓

b. $\frac{1}{17}$

c. 16

d. $\frac{17}{16}$

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/16^{n+1}}{n/16^n} = \frac{n+1}{16n} \rightarrow \frac{1}{16}$. Since $\frac{1}{16} < 1$, the series converges by the Ratio Test.

32 Use the Ratio Test to find $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ for $\sum_{n=1}^{\infty} \frac{n}{6^n}$.

a. $\frac{1}{7}$

b. $\frac{1}{6}$ ✓

c. $\frac{7}{6}$

d. 6

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/6^{n+1}}{n/6^n} = \frac{n+1}{6n} \rightarrow \frac{1}{6}$. Since $\frac{1}{6} < 1$, the series converges by the Ratio Test.

Multiple choice — medium 15

33 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{4n+4}{11n-7} \right)^n \right|}$.

a. $\frac{4}{11}$ ✓

b. $\frac{11}{4}$

c. $\frac{19}{22}$

d. $\frac{15}{11}$

$\sqrt[n]{\left(\frac{4n+4}{11n-7} \right)^n} = \frac{4n+4}{11n-7} \rightarrow \frac{4}{11} = \frac{4}{11}$. Since $\frac{4}{11} < 1$, converges by the Root Test.

34 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{5n+8}{10n+3} \right)^n \right|}$.

a. 1

b. 2

c. $\frac{1}{2}$ ✓

d. $\frac{3}{2}$

$\sqrt[n]{\left(\frac{5n+8}{10n+3} \right)^n} = \frac{5n+8}{10n+3} \rightarrow \frac{5}{10} = \frac{1}{2}$. Since $\frac{1}{2} < 1$, converges by the Root Test.

35 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n-7}{12n+4} \right)^n \right|}$.

a. $\frac{12}{7}$

b. $\frac{7}{12}$ ✓

c. $\frac{19}{12}$

d. $\frac{13}{12}$

$\sqrt[n]{\left(\frac{7n-7}{12n+4} \right)^n} = \frac{7n-7}{12n+4} \rightarrow \frac{7}{12} = \frac{7}{12}$. Since $\frac{7}{12} < 1$, converges by the Root Test.

36 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{3n+4}{8n-6} \right)^n \right|}$.

a. $\frac{7}{8}$

b. $\frac{8}{3}$

c. $\frac{3}{8}$ ✓

d. $\frac{11}{8}$

$\sqrt[n]{\left(\frac{3n+4}{8n-6} \right)^n} = \frac{3n+4}{8n-6} \rightarrow \frac{3}{8} = \frac{3}{8}$. Since $\frac{3}{8} < 1$, converges by the Root Test.

37 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{5n+7}{11n+4} \right)^n \right|}$.

a. $\frac{5}{11}$ ✓

b. $\frac{11}{5}$

c. $\frac{21}{22}$

d. $\frac{16}{11}$

$\sqrt[n]{\left(\frac{5n+7}{11n+4} \right)^n} = \frac{5n+7}{11n+4} \rightarrow \frac{5}{11} = \frac{5}{11}$. Since $\frac{5}{11} < 1$, converges by the Root Test.

38 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{4n+6}{9n-2} \right)^n \right|}$.

a. $\frac{9}{4}$

b. $\frac{4}{9}$ ✓

c. $\frac{17}{18}$

d. $\frac{13}{9}$

$\sqrt[n]{\left(\frac{4n+6}{9n-2} \right)^n} = \frac{4n+6}{9n-2} \rightarrow \frac{4}{9} = \frac{4}{9}$. Since $\frac{4}{9} < 1$, converges by the Root Test.

39 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{2n-1}{11n-2} \right)^n \right|}$.

a. $\frac{2}{11}$ ✓

b. $\frac{15}{22}$

c. $\frac{11}{2}$

d. $\frac{13}{11}$

$\sqrt[n]{\left(\frac{2n-1}{11n-2} \right)^n} = \frac{2n-1}{11n-2} \rightarrow \frac{2}{11} = \frac{2}{11}$. Since $\frac{2}{11} < 1$, converges by the Root Test.

40 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{9n+4}{12n+3} \right)^n \right|}$.

a. $\frac{3}{4}$ ✓

b. $\frac{7}{4}$

c. $\frac{4}{3}$

d. $\frac{5}{4}$

$\sqrt[n]{\left(\frac{9n+4}{12n+3} \right)^n} = \frac{9n+4}{12n+3} \rightarrow \frac{9}{12} = \frac{3}{4}$. Since $\frac{3}{4} < 1$, converges by the Root Test.

41 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{n+3}{7n+8} \right)^n \right|}$.

a. $\frac{9}{14}$

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{7}$ ✓

$\sqrt[n]{\left(\frac{n+3}{7n+8} \right)^n} = \frac{n+3}{7n+8} \rightarrow \frac{1}{7} = \frac{1}{7}$. Since $\frac{1}{7} < 1$, converges by the Root Test.

42 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{6n-6}{12n-3} \right)^n \right|}$.

a. $\frac{3}{2}$

b. $\frac{1}{2}$ ✓

c. 2

d. 1

$\sqrt[n]{\left(\frac{6n-6}{12n-3} \right)^n} = \frac{6n-6}{12n-3} \rightarrow \frac{6}{12} = \frac{1}{2}$. Since $\frac{1}{2} < 1$, converges by the Root Test.

43 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n-7}{16n+2} \right)^n \right|}$.

a. $\frac{7}{16}$ ✓

b. $\frac{16}{7}$

c. $\frac{15}{16}$

d. $\frac{23}{16}$

$\sqrt[n]{\left(\frac{7n-7}{16n+2} \right)^n} = \frac{7n-7}{16n+2} \rightarrow \frac{7}{16} = \frac{7}{16}$. Since $\frac{7}{16} < 1$, converges by the Root Test.

44 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{8n-6}{13n+7} \right)^n \right|}$.

a. $\frac{21}{13}$

b. $\frac{8}{13}$ ✓

c. $\frac{29}{26}$

d. $\frac{13}{8}$

$\sqrt[n]{\left(\frac{8n-6}{13n+7} \right)^n} = \frac{8n-6}{13n+7} \rightarrow \frac{8}{13} = \frac{8}{13}$. Since $\frac{8}{13} < 1$, converges by the Root Test.

45 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{6n-6}{8n-2} \right)^n \right|}$.

a. $\frac{3}{4}$ ✓

b. $\frac{4}{3}$

c. $\frac{7}{4}$

d. $\frac{5}{4}$

$\sqrt[n]{\left(\frac{6n-6}{8n-2} \right)^n} = \frac{6n-6}{8n-2} \rightarrow \frac{6}{8} = \frac{3}{4}$. Since $\frac{3}{4} < 1$, converges by the Root Test.

46 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{7n+6}{11n-9} \right)^n \right|}$.

a. $\frac{25}{22}$

b. $\frac{11}{7}$

c. $\frac{18}{11}$

d. $\frac{7}{11}$ ✓

$\sqrt[n]{\left(\frac{7n+6}{11n-9} \right)^n} = \frac{7n+6}{11n-9} \rightarrow \frac{7}{11} = \frac{7}{11}$. Since $\frac{7}{11} < 1$, converges by the Root Test.

47 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{8n-5}{12n-4} \right)^n \right|}$.

a. $\frac{3}{2}$

b. $\frac{2}{3}$ ✓

c. $\frac{5}{3}$

d. $\frac{7}{6}$

$\sqrt[n]{\left(\frac{8n-5}{12n-4} \right)^n} = \frac{8n-5}{12n-4} \rightarrow \frac{8}{12} = \frac{2}{3}$. Since $\frac{2}{3} < 1$, converges by the Root Test.

Multiple choice — hard 15

48 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 5}{n^1}$ converges absolutely, conditionally, or diverges.

a. Diverges

b. Cannot be determined without more information

c. **Conditionally convergent** ✓

d. Absolutely convergent

The series alternates and $b_n = \frac{5}{n^1} \rightarrow 0$, decreasing, so it converges by the AST. Testing absolute convergence: since $\sum \left| \frac{5}{n^1} \right| = \sum \frac{5}{n^1}$ diverges ($p = 1$). So the series is **CONDITIONALLY CONVERGENT**.

49 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+6}$ converges absolutely, conditionally, or diverges.

a. Cannot be determined without more information

b. **Diverges** ✓

c. Conditionally convergent

d. Absolutely convergent

$b_n = \frac{n}{n+6} \rightarrow 1 \neq 0$. Since the terms don't approach 0, the series diverges by the n th-term test.

50 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 3}{n^3}$ converges absolutely, conditionally, or diverges.

- a. Conditionally convergent
- b. Cannot be determined without more information
- c. Diverges
- d. **Absolutely convergent ✓**

The series alternates and $b_n = \frac{3}{n^3} \rightarrow 0$, decreasing, so it converges by the AST. Testing absolute convergence: since $\sum \left| \frac{3}{n^3} \right| = \sum \frac{3}{n^3}$ converges ($p = 3$). So the series is **ABSOLUTELY CONVERGENT**.

51 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 4}{n^1}$ converges absolutely, conditionally, or diverges.

- a. Absolutely convergent
- b. Cannot be determined without more information
- c. Diverges
- d. **Conditionally convergent ✓**

The series alternates and $b_n = \frac{4}{n^1} \rightarrow 0$, decreasing, so it converges by the AST. Testing absolute convergence: since $\sum \left| \frac{4}{n^1} \right| = \sum \frac{4}{n^1}$ diverges ($p = 1$). So the series is **CONDITIONALLY CONVERGENT**.

52 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n 7}{n^{\frac{3}{2}}}$ converges absolutely, conditionally, or diverges.

- a. Diverges
- b. Cannot be determined without more information
- c. Conditionally convergent
- d. **Absolutely convergent ✓**

The series alternates and $b_n = \frac{7}{n^{\frac{3}{2}}} \rightarrow 0$, decreasing, so it converges by the AST. Testing absolute convergence: since $\sum \left| \frac{7}{n^{\frac{3}{2}}} \right| = \sum \frac{7}{n^{\frac{3}{2}}}$ converges ($p = \frac{3}{2}$). So the series is **ABSOLUTELY CONVERGENT**.

53 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{8^n}{n!}$ converges.

- a. Diverges ($L = \infty$)
- b. Converges by the Alternating Series Test
- c. Inconclusive ($L = 1$)
- d. Converges ($L = 0$) ✓

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{8^{n+1}/(n+1)!}{8^n/n!} = \frac{8}{n+1} \rightarrow 0. \text{ Converges by the Ratio Test.}$$

54 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

- a. Inconclusive ($L = 1$)
- b. Converges by the Alternating Series Test
- c. Diverges ($L = \infty$)
- d. Converges ($L = 0$) ✓

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3^{n+1}/(n+1)!}{3^n/n!} = \frac{3}{n+1} \rightarrow 0. \text{ Converges by the Ratio Test.}$$

55 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{6^n}$ converges.

- a. Converges ($L = 0$)
- b. Converges by the Alternating Series Test
- c. Diverges ($L = \infty$) ✓
- d. Inconclusive ($L = 1$)

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!/6^{n+1}}{n!/6^n} = \frac{n+1}{6} \rightarrow \infty. \text{ Diverges by the Ratio Test.}$$

56 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{2^n}$ converges.

- a. Diverges ($L = \infty$) ✓
- b. Converges by the Alternating Series Test
- c. Inconclusive ($L = 1$)
- d. Converges ($L = 0$)

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!/2^{n+1}}{n!/2^n} = \frac{n+1}{2} \rightarrow \infty. \text{ Diverges by the Ratio Test.}$$

57 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{7^n}{n!}$ converges.

a. Inconclusive ($L = 1$)

b. Converges ($L = 0$) ✓

c. Diverges ($L = \infty$)

d. Converges by the Alternating Series Test

$\left| \frac{a_{n+1}}{a_n} \right| = \frac{7^{n+1}/(n+1)!}{7^n/n!} = \frac{7}{n+1} \rightarrow 0$. Converges by the Ratio Test.

58 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n}{n+3}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+3}\right)^n$.

a. 3

b. 9 ✓

c. 8

d. 10

$\sqrt[n]{\left(\frac{9n}{n+3}\right)^n} = \frac{9n}{n+3} \rightarrow 9$ as $n \rightarrow \infty$. Since $9 > 1$, diverges by the Root Test.

59 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{3n}{n+5}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{3n}{n+5}\right)^n$.

a. 4

b. 2

c. 5

d. 3 ✓

$\sqrt[n]{\left(\frac{3n}{n+5}\right)^n} = \frac{3n}{n+5} \rightarrow 3$ as $n \rightarrow \infty$. Since $3 > 1$, diverges by the Root Test.

60 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{5n}{n+3}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{5n}{n+3}\right)^n$.

a. 4

b. 5 ✓

c. 3

d. 6

$\sqrt[n]{\left(\frac{5n}{n+3}\right)^n} = \frac{5n}{n+3} \rightarrow 5$ as $n \rightarrow \infty$. Since $5 > 1$, diverges by the Root Test.

61 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{9n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{9n}{n+1}\right)^n$.

a. 1

b. 8

c. 10

d. 9 ✓

$\sqrt[n]{\left(\frac{9n}{n+1}\right)^n} = \frac{9n}{n+1} \rightarrow 9$ as $n \rightarrow \infty$. Since $9 > 1$, diverges by the Root Test.

62 Use the Root Test to find $L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{5n}{n+1}\right)^n}$ for $\sum_{n=1}^{\infty} \left(\frac{5n}{n+1}\right)^n$.

a. 5 ✓

b. 6

c. 4

d. 1

$\sqrt[n]{\left(\frac{5n}{n+1}\right)^n} = \frac{5n}{n+1} \rightarrow 5$ as $n \rightarrow \infty$. Since $5 > 1$, diverges by the Root Test.